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All That Is Urban Melts into Data: Circulations of Matter, Energy, and Data in the Digital City

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ABSTRACT

This article seeks to conceptualize the interrelated movement of data on the one hand, and matter and energy on the other hand—and therefore contributes to a theorization of data as simultaneously constitutive of, and constituted by, contemporary urbanism. To pursue this objective, we argue for close dialogue between urban political ecology (UPE) on the one hand, and science and technology studies (STS) and critical data studies (CDS) on the other hand. More specifically, we call for a closer examination of how data move, accumulate and agglomerate, and we propose that a conceptual distinction between “flows” and “circulations” can help operationalize this research agenda. We contend that this dialogue opens exciting research perspectives for both fields. From a UPE perspective, paying closer attention to data circulations is a way to advance poststructuralist approaches because it allows for fine-grained analysis of how resources (data, matter, and energy) are circulated, transformed, and accumulated to produce new urban natures. From an STS and CDS perspective, adopting a metabolic perspective on data circulations helps to reconceptualize the urban and analyze cities as sites of data agglomeration, accumulation, and contestation.

KEYWORDS

urbanizing science and technology studies; digital materialism; circulations and flows; urban metabolism; digital urbanism

Introduction

Vignette #1: Data Itineraries

In recent years, a few scholars have organized “data walks” intended to extend conversations around processes of data production/collection, aggregation, valuation and usage, by adopting a street-level perspectives on the power dynamics at play (Powell, 2018; Mahmoudi et al., 2024). The idea of such walks is to document interactions with the digital that occur as people use urban space and explore how these data then feed into a political economy of digital surveillance (Zuboff, 2019). Consider how this line of thinking could be expanded upon by exploring the creation of these data, *physically*. The exercise is highly speculative in nature, given the opacity of tech firms’ infrastructures and daily operations, but it could look as follows. Upon creating a data point, the data walk participant’s smartphone may send information to an edge data center—a smaller type of data center

located in denser urban settings so as to reduce latency and allow for faster service. The data are then duplicated (for the added security that redundancy offers) and sent to several larger data centers across the continent. It may then be sorted, agglomerated, and sold to a third-party company that will broker a deal with another company eager to target the participant with a new online marketing campaign.

Vignette #2: The Crypto-Mining Facility

Residents of Raimbault Street in Sherbrooke, Québec, are blessed with backyards that overlook the Magog River. In 2020, however, a continuous hum started tainting their experience as they spent time in their backyards, enjoying the pastoral view. The source of this nuisance was found across the river, at the edge of a small industrial zone. In a former warehouse, a Toronto-based company named Bitfarms had started operating a Bitcoin mining facility—one of three located in Sherbrooke. Residents' discontent drew attention to an industry that started flourishing in the Canadian province of Québec, where hydropower plants generate cheap electricity. The momentum that cryptocurrency mining was gaining was abruptly halted as the provincial energy regulator put a moratorium on new mining facilities in Québec, and this particular facility was later relocated to a larger industrial district where it no longer bothers residents. Nevertheless, three of Bitfarms' data centers are currently operating in Sherbrooke, consuming over 100mw of power, which makes them, by far, the largest electricity consumer in the city.

The materiality of data highlighted by these two vignettes constitutes the starting point of the present article. Vignette #1 shows that while our interactions with the digital usually occur between the user and a terminal, a vast infrastructure and an array of actors allow for the circulation of data. Meanwhile, with vignette #2, we seek to emphasize that the circulation of data is also contingent on the circulation of other resources—here, electricity flows to the Bitfarms facility, where it is transformed into Bitcoins and residual heat. In recent years, scholars have indeed turned their attention more to the “material” dimensions of digital urban lives (Lockhart et al., 2023; Lupton, 2019; Turnbull et al., 2019). Among their many implications, it is becoming harder to ignore the environmental costs associated with our everyday use of digital devices—from the materials involved in the construction of hardware like phones and computers (Cubitt, 2016), to the energy consumption of software, and notably cloud-based solutions (Ipsen et al., 2019; McLean, 2023). The realization that the digital realm is subtended by nefarious extractive practices (Pitron, 2021; Arboleda, 2020) and insatiable large-scale energy systems (Lopez, 2023; Sadowski and Levenda, 2020) has slowly put the (in)visibility of the Internet's infrastructure on geographers' agenda (Furlong, 2021). From a theoretical standpoint, this attention to the “matter” of the Internet opens new perspectives for apprehending how the social world is represented and constituted, as noted by Bakker and Bridge (2006: 18):

Referencing “the material” is thus a way of acknowledging the embeddedness of social action, whether it be embedded in world of things, bodies, networks or socio-ecological relations ... Yet “the material” is more than just a call for a heterogeneously populated world: it is also an acknowledgement that the “things” (commodities, bodies, biophysical processes) that make a difference in the way social relations unfold are not pregiven substrates that variably enable and constrain social action, but are themselves historical products of material, representational, and symbolic practices.

In this article, we build upon this reflection on the materiality of the digital and focus on how the circulation of data shapes urban space. This urban focus is because contemporary cities are marked by concentrated—and heightened circulation of—natural, human, and digital resources. However, the means by which these resources accumulate, organize, and flow in and through cities remains underexplored in Science and Technology Studies (STS), a point recently made by Coutard and Florentin (2022) and by Smith and Martin (2021). A lot remains to be explored as we seek to understand new socio-technical entanglements made up of data, infrastructures, apps, or capital, which create and constrain movement of information, matter, and energy. To advance this research agenda, we put forward a conceptual distinction between *flows* and *circulations*. By operating this distinction, we contend that we can improve dialogue between STS and Critical Data Studies (CDS) on the one hand, and Urban Political Ecology (UPE) on the other hand, thus enriching both fields. More specifically, as we expand on how to think about flows and circulations in urban space, we concurrently seek to:

- extend ongoing debates in UPE by providing a conceptual tool that can strengthen the “poststructuralist” turn in this field
- “urbanize” STS and CDS by showing how cities are not merely a stage for data circulations, but rather an actor in its own right.

To carry out these objectives and advance conversations between CDS and UPE, a first step is to integrate streams of data to the concept of *urban metabolism*: the next section will pursue this objective by offering an examination of where data “fit” in Marxist understandings of urbanization processes, then fleshing out the distinction we propose between flows and circulations, and finally offering an operationalization of our conceptual framework with an account of new ways to think of digital geographies. The following section will then highlight the contributions of the data flows/circulations distinction for UPE by focusing on the relationship between data, matter, and energy. Finally, the last section will address the contributions that the same conceptual distinction can offer to rethink CDS and give them a stronger urban dimension.

Integrating Data into Urban Metabolism

To establish closer dialogue between the fields of UPE and CDS, we find that the concept of *urban metabolism* offers promising perspectives. This section will explore how we can integrate data into this conceptual framework. Of note, we follow Flensburg and Lai (2023: 321) in our understanding of data:

“Data” is thus defined as the packages that all types of digital content are broken down to when transported over the Internet—regardless of whether they contain media content in a conventional sense and flow from content providers to individual users, or if they flow from users to content providers carrying meta-data about the users’ online behavior, browser history, location, and other valuable information.

With this understanding of data as units that circulate and accumulate across digital infrastructures, the section seeks to explore the possibility of updating a Marxist perspective on the urbanization of nature in the digital age and then outlines the distinction that

can be made between *flows* and *circulations* of data, before outlining the advantages of this overall framework for rethinking digital geographies.

Data and the Urbanization of Nature

UPE theorizes the co-production of society and nature by analyzing distributions of diverse resources in urban environments (Heynen et al., 2006; Keil, 2003). To advance this agenda, UPE has heavily relied on two concepts: *circulation* and *metabolism*. Swyngedouw (2006) provides a detailed account of the origins and subsequent mobilizations of these concepts. To summarize these elements, Marx was one of the earliest adopters of the term “metabolism,” which had been coined in the early nineteenth century to designate “the exchange of energy and substances between organisms and the environment on the one hand and the totality of biochemical reactions in a living thing on the other” (Swyngedouw, 2006: 107). For him, metabolism offered a new perspective to think about socio-environmental change. More precisely, he used it as a metaphor to develop his definition of labor and conceptualize the relationship between human societies and nature:

Labor is, first of all, a process between man and nature, a process by which man, through his own actions, mediates, regulates, and controls the *metabolism* between himself and nature ... Through this movement he acts upon external nature and changes it, and in this way, he simultaneously changes his own nature ... (Marx, 1970: 283, 290)

Human labor, in other words, effectuates the movement, exchange, and transformation of matter that makes up (socio-natural) urban metabolism. Here enters the notion of “circulation.” Swyngedouw (2006: 107) notes that its scientific use gained traction alongside the notion of metabolism, and that while it was initially used to refer to the movement of blood through the human vascular system, it was quickly adopted by seventeenth-century thinkers to designate the movements of money through a given national economy. The metaphor became well-entrenched in Western political economic tradition, with the notion of circulation acquiring an extremely positive connotation:

Adam Smith and, in particular, Karl Marx conceived a capitalist economy as a metabolic system of circulating money and commodities, carried by and structured through social interactions and relations. Accumulation is dependent on the swiftness by which money circulates through society. Each hiccup, stagnation, or interruption of circulation may unleash the infernal forces of devaluation, crisis, and chaos. (Swyngedouw, 2006: 111)

How do these concepts apply to data in the digital era? First, they underscore that the circulation of data is a consequence of human labor (Denis, 2018; Fuchs, 2016; Manzerolle and Kjosén, 2017). The idea is not new: in the 1970s, Dorothy Smith pointed out that the reality we experience in our everyday life hinges upon a multiplicity of written accounts, which are the result of work and work organization. Things that are as basic as issuing a birth certificate result from specific professional practices and answer distinct objectives, thereby explaining why some data are included in the certificate and not other data (Smith, 1974). In the twenty-first century, these data are circulated digitally. In spite of a common misconception that they are handled automatically by way of ever-sophisticated algorithms, their circulation requires considerable human work (Fuchs, 2016; Zheng and Sieber, 2021). Indeed, a large literature has emerged over more than a

decade to theorize the diverse ways in which humans' digitally mediated work produces, formats, categorizes, curates, circulates, directs, and consumes data and value (for reviews and critical analyses of this literature, see Jarrett, 2022 and Scholz, 2012). A Marxian perspective on urban metabolism thus helps us think about how data circulate through labor, and therefore ultimately, through social relations (Burns and Welker, 2022).

Second, UPE's conceptualization of circulations is directed towards a better understanding of resource distribution. In other words, metabolic flows result from power struggles and, therefore, allow us to read power relations, social tensions, and ultimately socio-spatial inequalities and injustice (Cook and Swyngedouw, 2012). UPE invites us to look at where data flow and accumulate—in other words, it prompts us to look not only at flows, but also at “stocks” of data. In doing so, data extractivism practices can be analyzed as practices that perpetuate and reinforce social relegation (Rosol and Blue, 2022) or neo-colonial relations (Mouton and Burns, 2021) in urban space. This line of thought is especially valuable as data have become a “structural condition of capital itself” (Halpern et al., 2022: 199), meaning that their historical nature as pieces of information that describe and represent the world is eclipsed by their new function as generators of value. For Halpern et al. (2022), this extension of the role of data (beyond their mere descriptive use) echoes Marx's definition of “surplus labor.” Just as “labor confers value on the materials it transforms” and becomes *surplus* when it “operates to build capital rather than replace it,” data become *surplus data* when their quantity offers promises of never-ending valuation.

Flows vs. Circulations

Up until this point, we have used the notions of “flow” and “circulation” interchangeably. To pursue our exploration of how data streams co-evolve with urbanization processes; however, we need to clarify our terminology. To this end, humor us by undertaking a thought experiment. You construct a small origami paper boat and set it afloat on a swift stream. The boat moves in some predictable ways—it moves downstream, it orients in the general direction of the water flow—but mostly behaves in apparent chaos, guided left or right by micro-forces that are beyond your perceptive capabilities. To describe this object's behavior as “circulation” would be incorrect, because the term implies a degree of cohesion or logic, a circularity and potential of intentionality. Instead, the boat “flows” along the stream from an origin point to a downstream location. The water in the stream *circulates* because it is interlinked in broader hydrological systems and, to our epistemological frames, can seem to emanate a logic. Building from this analogy, and borrowing from Garcier et al. (2017), as well as Blaszkiewicz (2023), the notion of information emerges as a critical notion to distinguish flows and circulations: “Information on flows correspond to an aggregate, for which individual characteristics have been lost, and where exact space-time progression is not known with a high degree of precision” (Bahoken et al., 2016: 65, cited in Garcier et al., 2017, author's translation).

As detailed in Table 1, focusing on circulations rather than flows involves using a new lens—a macro one—which allows for movements of matter and energy to be disaggregated and studied more closely. Among the myriads of scholarly works that mobilize the notion of “metabolism,” it brings us further away from perspectives that rely on

Table 1: Data flows vs. data circulations

	Flows	Circulations
Focus	Point of origin and destination	Trajectory and transformations
Analytical Lens	Quantitative (e.g., volume of data)	Qualitative (ethnographic and descriptive)

quantitative tools like mass flow analysis, and closer to fine-grained descriptions and analyses of local practices that involve the use and transformation of resources (Desvaux, 2019, 2024).

From a practical perspective, what does studying *circulations* rather than *flows* look like as we try to operationalize the research agenda outlined earlier in this section and adopt a UPE perspective on data? This means that we are not as concerned about quantifying volumes of data that are being generated and received as we are interested in retracing the journey they take: how and by whom they are produced in the first place, which entities acquire and (re)sell them, how they are curated, collated and sorted, where they are stored, etc. In other words, much like social scientists have found heuristic value in *following* a variety of objects, from papaya (Cook 2004) and mushrooms (Tsing, 2015) to money (Christophers, 2011) to cement (Choplin, 2020) to data (Akbari, 2020; Flensburg and Lai, 2023), we advocate for a close examination of each step of data’s journey—from how, where and by whom they are produced to where they are stored, how they are collated, with what they are agglomerated, to whom they are sold, etc. According to Akbari (2020), “following” data circulations could also illuminate the discursive-material ways in which each confrontation and point of tension in data’s journey leads us to rethink data themselves, and the roles they play in reconfiguring urban space.

The methodological ramifications of this approach still need to be fleshed out and would merit a separate discussion. In the present article, however, a couple of elements can be highlighted. First, such inquiries are ethnographic in nature, and build upon a now well-established tradition of *digital ethnography* that considers how databases (Burns and Wark, 2019), algorithms (Seaver, 2017), or virtual spaces (Hine, 2017) contribute to an ordering of the social world they are embedded in. Second, following data involves careful attention to infrastructures (Flensburg and Lai 2023) and raises questions regarding how various components of data infrastructure reflect and (re)produce social meaning. As the next subsection will argue, it is a useful tool to expand and refine geographies of the digital.

Rethinking Geographies of the Digital

Thinking about data circulations is an invitation to rethink the geographies of the digital. If we follow the data, spaces of production first need to be considered. In this regard, Barns (2020) convincingly demonstrates that cities are a space of “ambient computing” whose density is essential to the flourishing of platforms and the scaling up of data production. This is especially true when tech companies take direct stakes in urban (re)development and shape urban space so that data production is maximized, and feedback loops of data are built into the urban fabric (Mahmoudi et al., 2024).

Next, spaces of data storage and computation can be highlighted. In this regard, scholars have initiated a fruitful conversation on the location of the physical infrastructure of

the Internet (Edwards et al., 2024; Mouton, 2024), from attempts at “mapping” data centers (Carr et al., 2022; Durairajan et al., 2013) to examinations of how these data storage facilities are embedded in and transform local settings. These explorations have taken the form of ethnographic inquiries into how data centers coexist with their urban environment (Marquet, 2019; Ortar et al., 2022), how they use up adjacent resources—energy (Diguët and Lopez, 2019; Libertson et al., 2021; Bresnihan and Brodie, 2023) or water (Hogan, 2015; Albornoz and Tironi, this issue)—or how they mobilize and reshape territorial imaginaries and identities (Johnson, 2019; Vonderau, 2017; Brodie and Velkova, 2021). In grappling with this relatively new research object, urban scholars have discussed the interplay between analog and digital geographies, showing that the accumulation and ever-increasing circulation of data have a material footprint that transform urbanization. On this note, as underlined by Pickren (2018), a UPE perspective on the geography of data storage highlights connections between cities and their hinterland—some of the data centers being located in urban areas and others in remote places, and both relying on electronic components assembled elsewhere, from minerals extracted in yet another location. Along the same lines, Levenda and Mahmoudi (2019) emphasize the extractive logics at play, as natural resources as mobilized for the reproduction of digital capitalism.

Finally, the geography of data stocks also includes spaces where data are transformed into capital or, in other words, areas where host companies engaged in data extraction practices. This focus has distinct implications for urban studies, because it turns our attention to a different set of spatial footprints: it has to do with how major actors of digital capitalism change the governance of urban areas (McNeill, 2016), disrupt land-ownership patterns and foster gentrification (Chapple, 2017).

The Data-Matter-Energy Nexus: New Perspectives for UPE

The distinction between circulations and flows sets the stage for thinking about how data affects streams of matter and energy. In other words, integrating data streams into UPE is not simply a call to view data circulations under a different (theoretical) light: we argue that it opens new perspectives for thinking about circulations of matter and energy more in general. A salient element in our discussion is indeed that data are simultaneously flowing and structuring other flows—remember the introductory vignette that presented the Bitcoin mining facility, which presents itself as a strong gravitation pull for electricity and initiates new circulations of cryptocurrency data. Here, we wish to go a step further and argue that data are turning streams of matter and energy from flows to circulations. This process results from increased integration between digital technologies and urban infrastructures. Such infrastructures (e.g., transportation, water and wastewater management, energy distribution, and waste collection and management) mediate the provisioning and distribution of resources for cities—what UPE scholars call the mediation of the city-nature relationship (Kaika, 2004; Swyngedouw, 2004). Data and urban infrastructures have always been closely associated—from how data on urban environments were collected and analyzed to design urban services (Chatzis, 1993) to how better circulation of information was needed to expand large technical systems (Morus, 2000), and how knowledge on the infrastructure itself is critical for its management and maintenance (Denis and Florentin, 2019). Digital technologies transformed the relationship

between data and infrastructures, however. With the gradual integration of data acquisition and processing technologies such as Supervisory Control And Data Acquisition (SCADA) or their more recent and urban-marketed iterations, urban operating systems (Luque-Ayala and Marvin, 2020), and digital twins (Sánchez-Vaquerizo, 2022), urban infrastructures are not only moving matter and energy; they are transforming them into data. Luque-Ayala and Rutherford (2023) highlight this datafication of ecological flows and further argue that this process increases the agency of infrastructures. To them, non-human elements such as infrastructures and ecological flows “find a voice” (2023: 71) through digital technologies, thus opening or widening a political arena for social intervention in the (re)production of cities. As flows of matter and energy become more visible, they become more governable (Moere and Hill, 2012; Van Dijck, 2014).

Apprehending urban metabolism through circulations rather than flows is a way to capture our increasing ability to visualize movements of matter and energy through the acquisition of new data. It also echoes ongoing developments in UPE, and particularly its “poststructuralist turn” (Cornea et al., 2017a; Lawhon et al., 2014) where scholars have complemented macro analyses of how major actors (e.g., platforms) intervene in the reshaping of urban environments with an examination of more diffuse practices of everyday city-making. This lens of analysis privileges more granular approaches and places the focus on situated practices of urbanization. Practically, this approach entails considering the transformation and circulation of resources and retracing their “metabolic pathways” (Desvaux, 2024) to understand how they change urban natures. In short, this strand of UPE advocates for a closer examination of the micropolitics at work in city-making, which invites new sets of methodological tools to be developed, that are ethnographic in nature (Cornea et al., 2017b). Here, we contend that furthering this research agenda can be achieved through a conjoint analysis of data and ecological circulations. While the datafication of urban services has been advocated by industry actors and city administrators as a panacea for turning cities into more efficient and more sustainable entities, its exact consequences for ecological circulations remain unclear. As many have noted, the “optimized” flows of urban phenomena is richly interwoven with politics, values, and epistemologies (Powell, 2021; Sadowski, 2020).

In pursuing this line of inquiry, other scholars (Hoefsloot et al., 2022; Mouton, 2024) have mobilized Scott’s (1998) notion of *legibility*. In *Seeing like a State*, Scott makes the argument that modern statecraft was built through a process of simplification. In order to exert control over their population, modern states did not seek to understand the complex social realities they confronted; rather, they sought to establish a few standards (e.g., naming practices, measurement systems, land cadasters) that enabled them to govern effectively. In other words, a few pieces of information were abstracted from the social world, and then used to establish a taxation mechanism, a justice and police apparatus, etc. As governments “narrowed their vision,” they also reshaped local communities through the creation of idealized visions of society, which were then imposed with the force of law. This gap between governments’ simplistic, techno-modernist visions and actually existing forms of social organization led, Scott argues, to “development fiascos,” where schemes intended to reorganize societies more efficiently and improve human livelihoods caused human tragedies, from economic breakdowns to armed conflicts between different segments of the population. This line of inquiry prompts us to consider

which data are being circulated—and, conversely, which aspects of ecological flows remain invisible to the eyes of city governments and service operators. This concept is also a provocation to interrogate the idea that the “big data” being disgorged into administrators’ systems provide a comprehensive view of urban life, and to remind us that data can only capture a fraction of social realities.

What is Urban about an Urbanized STS?

In the 2000’s, a number of scholars (e.g., Hommels, 2005; Coutard and Guy, 2007) lamented that the urban ramifications of STS remained vastly understudied. Since then, STS has been leveraged to explore the interplay between cities and technology—with the *Journal of Urban Technology* playing a prominent role pursuing this agenda, by analyzing how, for instance, the mutually constitutive nature of urban space and electricity (Verdeil and Jaglin, 2023) or infrastructure more broadly (Wiig et al., 2022). As we think about data circulations, we contend that beyond enriching urban studies with new perspectives or methodologies, STS can help us rethink our conceptualizations of the urban itself. To advance in this direction, we follow Wachsmuth’s (2012) historical lineage of urban metabolism to conceptualize the city as both a product, and an instigator, of flows and circulations. In particular, we want to revisit the longstanding debate about urban metabolism, from the direction of the Chicago School’s work on the city as an enclosed ecosystem to UPE’s reframing of the city as an object that is in dynamic relationality with “nature” (Wachsmuth, 2012). Doing so allows us to treat bytes, electrons, data, and other digital resources as materially equivalent to other “natural” resources such as data center-cooling water, silicon and lithium for smart device chips and batteries, and human resources for constructing and operating digital infrastructures. Effectively, this helps us to better understand what, exactly, is urban about circulations of digital-material objects.

Within the Chicago School tradition, Burgess (1925) is typically credited with most fully employing the urban metabolism metaphor. For Burgess, the city was a self-regulating organism in dualistic relationship with nature: the urban is the opposite of nature and is exclusive of it. For Burgess, the metabolism metaphor conceives human mobility across a city as *disorder*, and stasis as *order* (Wachsmuth, 2012). The latter of these is associated with demoralization, promiscuity, and vice. According to Gandy (2004, 364), “The value of the metabolic metaphor has been its emphasis on the complex interactions between social and bio-physical systems that allow the modern city to function...,” yet the analogy of a *body*’s metabolism renders the spatial relationships within cities into a functionalist analysis, where different spaces (e.g., a neighborhood, a planning zone, or a park) serve functions similar to organs in a body. Cutting across these functions, though, sparse attention is given to “nature,” or spaces Burgess might have seen to be “outside” the city.

In other words, Burgess’s influential conception of the city as comprised of circulations similar to a body had two fatal limitations: it was anthropocentric in that it prioritized the movement of humans, at the same time as it failed to recognize that the city is both dependent on, and indeed made through, relationships with “nature.” Instead, Gandy (2004: 364) asserts that “If we are to understand the contemporary dynamics of urban infrastructure, we need to overhaul the idea of urban metabolism inherited

from the nineteenth century in which functionalist conceptions of cities took precedence over the analysis of structural dimensions to urban change.” The point of “overhaul” here would be to preserve the utility of thinking in terms of urban circulations, while making more space for more-than-human actors, and deconstructing the problematic binary of nature-cities. Drawing on lessons from UPE discussed above, we are able to target these analyses of structural dimensions to observe the matrixed relations of more-than-human power in urban spaces, such that data and electrons are seen to circulate in the same way as water and air—that is to say, they are shaped by structural political and economic forces, rather than inert objects (see also Bennett, 2010; Whatmore, 2002).

Calling the urban “a multiscalar process of socio-spatial transformation” rather than a universal form, Brenner and Schmid (2015: 165) argue that the relations of urbanization have indeed become *planetary*, as former hinterlands have been reworked—*urbanized*—to facilitate extraction for the reformulated urban fabric. Arboleda (2020) has extended this analysis to show that the extraction subtending digital technological infrastructures and devices has also become planetary, extending its relations across a reworked globe, controlled and diverted through and by cities. In short, our efforts to urbanize STS and CDS need to acknowledge that there is nothing outside of urbanized relations within the digital, as urban relations increasingly instigate and shape flows and circulations of capital, human resources, natural resources, and digital resources such as data, transforming the hinterlands and placing them within a web of planetary logistics networks. A planetary urbanization framing, therefore, does not, contra some critique (e.g., McCarthy 2023), imply that rurality and the agrarian are unimportant to contemporary socio-political relations, but rather that they are increasingly brought into the fold of urbanization by exporting products to cities, by being remotely managed and controlled from cities, by utilizing methods of management and extraction from urban centers, and more. This diffusion of urban relations does not mean that the very notion of cities should be discarded, but instead that there are increasingly rarer phenomena that elude urban processes. Against this backdrop, cities can be understood as sites of accumulation, concentration, and contestation of flows and resources. A wide variety of theoretical frameworks acknowledge the role cities play in agglomerating and directing resources—from capital flows and mined resources that are converted into buildings (Graham, 2016; Harvey, 1978; Haila, 1988; Smith, 2010; Guironnet and Halbert, 2023) to ecological streams that are diverted to serve the interests of dominant social groups (Swyngedouw, 2004; Kaika, 2004) and even symbolic resources (e.g., cultural or social capital) that are accrued and converted (Wacquant, 2023).

Conclusion

Throughout this article, we have sought to establish closer dialogue between UPE and scholarship on digitalization, namely STS and CDS, with the intent to theorize how movements of data alter, and are altered by, movements of matter and energy. To explore this research avenue, we put forward a conceptual distinction that helped us consolidate our theoretical framework. We contrasted a focus on *flows*, where the key parameters are volume, point of origin and point of destination, with an analysis of *circulations*, which lends attention to specific (sometimes sinuous) trajectories and transformations. We argue that looking at circulations has two implications for UPE. First, it

encompasses a study of *data circulations* and opens new perspectives for thinking about urban metabolism, as we consider the joint movement of data and of the resources that are used and transformed to move, combine and store them. Second, as urban infrastructures are increasingly integrating digital technologies and producing data on the resources they carry (e.g., water, waste, heat, gas), a finer analysis of urban metabolism becomes possible—in other words, as matter and energy are increasingly more traceable, analyzing them in terms of circulations becomes more feasible. We, therefore, make a case for data circulations to contribute to UPE's poststructuralist turn. Meanwhile, we argued that a UPE understanding of data flows and circulations can help STS and CDS reframe the urban and give cities a clearer role, not as a mere backdrop of digitalization but as a crucial actor for the agglomeration, conversion, and at times contestation of digital resources.

Overall, a closer dialogue between UPE, STS and CDS can help advance discussions on the materiality of data. It does so by articulating two concurrent efforts towards apprehending said materiality: a focus on the infrastructures of data (Goldstein and Nost, 2022) and their ensuing material footprint, and an analysis of how urban environments are increasingly transformed by—and for—digitalization (Barns, 2022; Mahmoudi et al., 2024). While challenges still remain, notably from a methodological standpoint, we contend that taking a closer look at data circulations helps to materialize, urbanize, and politicize digitalization processes.

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Notes on Contributors

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References

- A. Akbari, "Follow the Thing: Data: Contestations over Data from the Global South," *Antipode* 52: 2 (2020) 408–429.
- M. Arboleda, *Planetary Mine: Territories of Extraction under Late Capitalism* (Brooklyn, NY: Verso Books, 2020).
- F. Bahoken, G. Claude, and C. Zanin, "D'une cartographie de flux à une cartographie du mouvement: Aspects sémiologiques," *Cartes and géomatique* 229–230 (2016) 65–74.
- K. Bakker, and G. Bridge, "Material Worlds? Resource Geographies and the Matter of Nature," *Progress in Human Geography* 30: 1 (2006) 5–27.
- S. Barns, "Re-Engineering the City: Platform Ecosystems and the Capture of Urban Big Data," *Frontiers in Sustainable Cities* 2: 32 (2020) 1–8.
- J. Bennett, *Vibrant Matter: A Political Ecology of Things* (Durham, NC: Duke University Press, 2010).
- N. Brenner, and C. Schmid, "Towards a New Epistemology of the Urban?" *City* 19: 2–3 (2015) 151–182.
- H. Blaszkiewicz, *Toujours plus vite? Logistique et capitalisme dans l'Afrique minière: Zambie/RD Congo* (Le Manuscrit, 2023).
- P. Bresnihan and P. Brodie, "Data Sinks, Carbon Services: Waste, Storage and Energy Cultures on Ireland's Peat Bogs," *New Media and Society* 25: 2 (2023) 361–383.
- P. Brodie and J. Velkova, "Cloud Ruins: Ericsson's Vaudreuil-Dorion Data Centre and Infrastructural Abandonment," *Information and Society* 24: 6 (2021) 869–885.
- R. Burns, and G. Wark, "Where's the Database in Digital Ethnography? Exploring Database Ethnography for Open Data Research," *Qualitative Research* 20: 5 (2020) 598–616.
- R. Burns, and P. Welker, "'Make Our Communities Better through Data': The Moral Economy of Smart City Labor," *Big Data and Society* 9: 1 (2022) 1–13.
- C. Carr, D. Bast, K. Madron, and A. Syrus, "Mapping the Clouds: The Matter of Data Centers," *Journal of Maps* 18: 1 (2022) 106–113.
- K. Chapple, "Income Inequality and Urban Displacement: The New Gentrification," *New Labor Forum* 26: 1 (2017) 84–93.
- K. Chatzis, "La régulation des systèmes socio-techniques sur la longue durée: le cas du système d'assainissement urbain," *Ecole Nationale des Ponts et Chaussées* (1993).
- A. Choplin, *Matière grise de l'urbain. La vie du ciment en Afrique* (Métis Presses, 2020).
- B. Christophers, "Follow the Thing: Money," *Environment and Planning D: Society and Space* 29: 6 (2011) 1068–1084.
- I. Cook, "Follow the Thing: Papaya," *Antipode* 36: 4 (2004) 642–664.
- I. Cook, and E. Swyngedouw, "Cities, Social Cohesion and the Environment: Towards a Future Research Agenda," *Urban Studies* 49: 9 (2012) 1959–1979.
- N. Cornea, R. Véron, and A. Zimmer, "Everyday Governance and Urban Environments: Towards a More Interdisciplinary Urban Political Ecology," *Geography Compass* 11: 4 (2017a) 1–12.
- N. Cornea, R. Véron, and A. Zimmer, "Clean City Politics: An Urban Political Ecology of Solid Waste in West Bengal, India," *Environment and Planning A* 49: 4 (2017b) 728–744.
- O. Coutard, and D. Florentin, "Resource Ecologies, Urban Metabolisms, and the Provision of Essential Services," *Journal of Urban Technology* 29: 1 (2022) 1–10.
- O. Coutard, and S. Guy, "STS and the City: Politics and Practices of Hope," *Science, Technology and Human Values* 32 (2007) 713–734.
- S. Cubitt, *Finite Media: The Environmental Implications of Digital Technologies* (Durham, NC: Duke University Press, 2016).
- J. Denis, *Le travail invisible des données: Éléments pour une sociologie des infrastructures scripturales* (Paris: Presses des Mines, 2018).
- J. Denis, and D. Florentin, "Visibility without Breakdown: Inquiries, Discoveries and Knowledge in the Maintenance of French Water Infrastructures," 4S Conference (2019).
- P. Desvaux, "Pour une approche qualitative du métabolisme urbain. L'exemple des voies métaboliques des déchets plastiques au Caire (Égypte)," *Flux* 2 (2019) 147–160.

- P. Desvaux, "Coding Urban Metabolism: Infrastructuring Metabolic Pathways," in O. Coutard and D. Florentin, eds., *Handbook of Infrastructures and Cities* (Northampton, MA: Edward Elgar, 2024) 353–365.
- C. Diguët, F. Lopez, and L. Lefèvre, "L'impact Spatial et Énergétique des Data Centers sur les Territoires," *ADEME, Direction Villes et Territoires Durables* (2019) 1–141.
- R. Durairajan, S. Ghosh, X. Tang, P. Barford, and B. Eriksson, "Internet Atlas: A Geographic Database of the Internet," *Proceedings of the 5th ACM workshop on HotPlanet* (2013).
- D. Edwards, Dustin, Z. Cooper, and M. Hogan, "The Making of Critical Data Center Studies," *Convergence* (2024). DOI:10.1177/13548565231224157
- S. Flensburg and S.S. Lai, "Follow the Data! A Strategy for Tracing Infrastructural Power," *Media and Communication* 11: 2 (2023) 319–329.
- C. Fuchs, "Marx's Capital in the Information Age," *Capital and Class* 41: 1 (2016) 51–67.
- K. Furlong, "Geographies of Infrastructure II: Concrete, Cloud and Layered (In)visibilities," *Progress in Human Geography* 45: 1 (2021) 190–198.
- M. Gandy, "Rethinking Urban Metabolism: Water, Space, and the Modern City," *City* 8: 3 (2004) 363–379.
- R. Garcier, E. Martinais, and L. Rocher, "Désigner, mesurer, réguler: la mise en politique des flux et circulations," *Géocarrefour* 91: 91/3 (2017).
- J. Goldstein and E. Nost, *The Nature of Data: Infrastructures, Environments, Politics* (Lincoln, NE: University of Nebraska Press, 2022).
- S. Graham, *Vertical: The City from Satellites to Bunkers* (Brooklyn, NY: Verso Books, 2016).
- A. Guironnet, and L. Halbert, *L'empire urbain de la finance* (Paris: Éditions Amsterdam, 2023).
- A. Haila, "Land as a Financial Asset: The Theory of Urban Rent as a Mirror of Economic Transformation," *Antipode* 20: 2 (1988) 79–101.
- O. Halpern, P. Jagoda, J. Kirkwood, and L. Weatherby, "Surplus Data: An Introduction," *Critical Inquiry* 48: 2 (2022) 197–210.
- D. Harvey, "The Urban Process Under Capitalism," *International Journal of Urban and Regional Research* 2: 1–3 (1978) 101–131.
- N. Heynen, M. Kaika, and E. Swyngedouw, *In the Nature of Cities: Urban Political Ecology and the Politics of Urban Metabolism* (London: Routledge, 2006).
- C. Hine, "From Virtual Ethnography to the Embedded, Embodied, Everyday Internet," in L. Hjorth, H. Horst, A. Galloway, G. Bell, eds., *The Routledge Companion to Digital Ethnography* (Abingdon: Routledge, 2017), 21–28.
- F. Hoefslot, C. Richter, J. Martínez, and K. Pfeffer, "The Datafication of Water Infrastructure and its Implications for (Il)legible Water Consumers," *Urban Geography* 44: 4 (2022) 1–23.
- M. Hogan, "Data Flows and Water Woes: The Utah Data Center," *Big Data and Society* 2: 2 (2015) 1–12.
- A. Hommels, "Studying Obduracy in the City: Toward a Productive Fusion between Technology Studies and Urban Studies," *Science, Technology, and Human Values* 30: 3 (2005) 323–351.
- K. Ipsen, R. Zimmermann, P. Nielsen, and M. Birkved, "Environmental Assessment of Smart City Solutions Using a Coupled Urban Metabolism—Life Cycle Impact Assessment Approach," *The International Journal of Life Cycle Assessment* 24 (2019) 1239–1253.
- K. Jarrett, *Digital Labor* (Boston: Polity Press, 2022).
- A. Johnson, "Emplacing Data within Imperial Histories: Imagining Iceland as Data Centers' 'Natural' Home," *Culture Machine* 18: 5 (2019) 1–12.
- M. Kaika, *City of Flows: Modernity, Nature, and the City* (London: Routledge, 2004).
- R. Keil, "Urban Political Ecology: Progress Report," *Urban Geography* 24: 8 (2003) 723–738.
- M. Lawhon, H. Ernstson, and J. Silver, "Provincializing Urban Political Ecology: Towards a Situated UPE through African Urbanism," *Antipode* 46: 2 (2014) 497–516.
- A. Levenda and D. Mahmoudi, "Silicon Forest and Server Farms: The (Urban) Nature of Digital Capitalism in the Pacific Northwest," *Culture Machine* 18 (2019) 1–14.
- F. Libertson, J. Velkova, and J. Palm, "Data-Center Infrastructure and Energy Gentrification: Perspectives from Sweden," *Sustainability: Science, Practice and Policy* 17: 1 (2021) 152–161.

- A. Lockhart, S. Marvin, and A. While, "Towards New Ecologies of Automation: Robotics and the Re-Engineering of Nature," *Geoforum* 145 (2023) 1–9.
- F. Lopez, *À Bout de Flux* (Quimperlé, France: Éditions divergences, 2023).
- D. Lupton, *Data Selves: More-Than-Human Perspectives* (Boston: Polity Press, 2019).
- A. Luque-Ayala, and S. Marvin, *Urban Operating Systems: Producing the Computational City* (Cambridge, MA: MIT Press, 2020).
- A. Luque-Ayala, and J. Rutherford, "Emerging Techno-Ecologies of Energy: Examining Digital Interventions and Engagements with Urban Infrastructure," in A. Wiig, K. Ward, T. Enright, M. Hodson, H. Pearsall, and J. Silver, eds., *Infrastructuring Urban Futures* (Bristol: Bristol University Press, 2023) 67–84.
- D. Mahmoudi, A. Levenda, and A. Sabatino, "The Urban-Tech Feedback Loop: A Surveillance and Development Data-Walk in South Lake Union," *Digital Geography and Society* 7 (2024): 100106.
- V. Manzerolle, and A.M. Kjosén, "Digital Media and Capital's Logic of Acceleration," in C. Fuchs, and V. Mosco, eds., *Marx in the Age of Digital Capitalism* (Chicago: Haymarket Books, 2017) 151–179.
- C. Marquet, "Binaire béton: Quand les infrastructures numériques aménagent la ville," Université Paris-Saclay (ComUE, 2019).
- K. Marx, *Capital* Volume 1 (London: Lawrence & Wishart Ltd, 1970).
- J. McCarthy, "Thoughts on 'Planetary Rural Geographies': Commonalities of Capitalist Development, Authoritarian Populisms, and energy transition," *Dialogues in Human Geography* (2023).
- S. McLean, "The Environmental Impact of ChatGPT: A Call for Sustainable Practices in AI Development," *Earth.org Energy* (2023) <https://earth.org/environmental-impact-chatgpt/> Accessed March 15, 2025.
- D. McNeill, "Governing a City of Unicorns: Technology Capital and the Urban Politics of San Francisco," *Urban Geography* 37 (2016) 494–513.
- A. Moere, and D. Hill, "Designing for the Situated and Public Visualization of Urban Data," *Journal of Urban Technology* 19: 2 (2012) 25–46.
- I. Morus, "'The Nervous System of Britain': Space, Time and the Electric Telegraph in the Victorian Age," *The British Journal for the History of Science* 33: 4 (2000) 455–475.
- M. Mouton, "Seeing Like an Urban Service Operator: Making Urban Circulations of Matter and Energy Legible in the Digital Age," in O. Coutard, and D. Florentine, eds., *Handbook of Infrastructures and Cities* (Northampton, MA: Edward Elgar Publishing, 2024) 340–352.
- M. Mouton, and R. Burns, "(Digital) Neo-Colonialism in the Smart City," *Regional Studies* 55: 12 (2021) 1890–1901.
- N. Ortar, J. Velkova, A. Taylor, P. Brodie, C. Marquet, A. Johnson, A. Pollio, and L. Cirolia, "Powering 'Smart' Futures: Data Centers and the Energy Politics of Digitalisation," *Energy Futures: Anthropocene Challenges, Emerging Technologies and Everyday Life* 10: 125 (2022) 125–168.
- G. Pickren, "'The Global Assemblage of Digital Flow': Critical Data Studies and the Infrastructures of Computing," *Progress in Human Geography* 42: 2 (2018) 225–243.
- G. Pitron, *L'enfer numérique: voyage au bout d'un like* (Paris: Éditions Les Liens qui libèrent, 2021).
- A. Powell, "The Data Walkshop and Radical Bottom-Up Data Knowledge," in *Ethnography for a Data-Saturated World* H. Knox and D. Nafus, eds. (Manchester: Manchester University Press, 2018), 212–232.
- A. Powell, *Undoing Optimization: Civic Action in Smart Cities* (New Haven, CT: Yale University Press, 2021).
- M. Rosol, and G. Blue, "From the Smart City to Urban Justice in a Digital Age," *City* 26: 4 (2022) 684–705.
- J. Sadowski, *Too Smart: How Digital Capitalism Is Extracting Data, Controlling Our Lives, and Taking over the World* (Cambridge, MA: MIT Press, 2020).
- J. Sadowski, and A. Levenda, "The Anti-Politics of Smart Energy Regimes," *Political Geography* 81 (2020) 1–8.

- J. Sánchez-Vaquerizo, "Getting Real: The Challenge of Building and Validating a Large-Scale Digital Twin of Barcelona's Traffic with Empirical Data," *ISPRS International Journal of Geo-Information* 11: 1 (2022) 1–28.
- T. Scholz, *Digital Labor: The Internet as Playground and Factory* (Abingdon: Routledge, 2012).
- J. Scott, *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed* (New Haven, CT: Yale University Press, 1998).
- N. Seaver, "Algorithms as Culture: Some Tactics for the Ethnography of Algorithmic Systems," *Big Data and Society* 4: 2 (2017) 1–12.
- A. Smith and P.P. Martín, "Going Beyond the Smart City? Implementing Technopolitical Platforms for Urban Democracy in Madrid and Barcelona," *Journal of Urban Technology* 28:1–2 (2021) 311–330.
- D. Smith, "The Social Construction of Documentary Reality," *Sociological Inquiry* 44: 4 (1974) 257–268.
- N. Smith, *Uneven Development: Nature, Capital, and the Production of Space* (3rd Edition) (Athens, GA: The University of Georgia Press, 2010).
- E. Swyngedouw, *Social Power and the Urbanization of Water: Flows of Power* (Oxford: Oxford University Press, 2004).
- E. Swyngedouw, "Circulations and Metabolisms: (Hybrid) Natures and (Cyborg) Cities," *Science as Culture* 15: 2 (2006) 105–121.
- A. Tsing, *The Mushroom at the End of the World: On the Possibility of Life in Capitalist Ruins* (Princeton, NJ: Princeton University Press, 2015).
- J. Turnbull, A. Searle, O. Davies, J. Dodsworth, P. Chasseray-Peraldi, E. von Essen, and H. Anderson-Elliott, "Digital Ecologies: Materialities, Encounters, Governance," *Progress in Environmental Geography* 2: 1–2 (2019) 3–32.
- J. Van Dijck, "Datafication, Dataism and Dataveillance: Big Data between Scientific Paradigm and Ideology," *Surveillance and Society* 12: 2 (2014) 197–208.
- E. Verdeil, and S. Jaglin, "Electrical Hybridizations in Cities of the South: From Heterogeneity to New Conceptualizations of Energy Transition," *Journal of Urban Technology* 30: 2 (2023) 1–10.
- A. Vonderau, "Technologies of Imagination: Locating the Cloud in Sweden's North," *Imaginations: Journal of Cross-Cultural Image Studies* 8: 2 (2017) 8–21.
- D. Wachsmuth, "Three Ecologies: Urban Metabolism and the Society-Nature Opposition," *The Sociological Quarterly* 53: 4 (2012) 506–523.
- L. Wacquant, *Bourdieu in the City: Challenging Urban Theory* (Hoboken, NJ: Polity Press, 2023).
- A. Wiig, A. Karvonen, C. McFarlane, and J. Rutherford, "Splintering Urbanism at 20: Mapping Trajectories of Research on Urban Infrastructures," *Journal of Urban Technology* 29: 1 (2022) 1–11.
- S. Whatmore, *Hybrid Geographies: Natures, Cultures, Spaces* (Thousand Oaks, CA: Sage, 2002).
- Z. Zheng, R. Sieber, "Putting Humans Back in the Loop of Machine Learning in Canadian Smart Cities," *Transactions in GIS* 26: 1 (2021) 8–24.
- S. Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (New York City: Public Affairs, 2019).